Post-doctoral researcher position :
Development of causal inference methods for survival data with a cure fraction

Duration : 12 months
Starting date : as soon as possible

Desbrest Institute of Epidemiology and Public Health, University of Montpellier, France

IDESP presentation

The Desbrest Institute of Epidemiology and Public Health (Institut Desbrest d'Epidémiologie et de Santé Publique – IDESP) is a joint French research unit between the University of Montpellier and Inserm aiming to understand the development of chronic diseases that affect more than 30% of the French population and represent 80% of healthcare expenditures. It consists of 83 members, among which 29 are qualified to supervise research. It hosts researchers from Inserm, Inria, the University of Montpellier, Montpellier and Nîmes University hospitals, Montpellier Cancer Institute and the university department of general practice of Montpellier. One of its research areas is statistical and mathematical modelling of patient trajectories.

Context

Survival analysis relies on the assumption that all subjects under study will eventually experience the event of interest. In some contexts, however, this assumption is not fulfilled. This is for example the case when interest lies in the time-to-relapse for early-stage breast cancer, or in the time to re-intubation for patients in an intensive care unit. In these two contexts, a fraction of patients may never experience a relapse and can therefore be considered long-term survivors or cured. Survival data is then a mixture between two types of observations: cured and uncured.

An important issue when dealing with survival analysis is the presence of censoring and particularly of right-censoring implying that the event of interest may not be observed for all observations due to losses to follow-up or the end of the study. Since cured patients never experience the event, it is not possible, a priori, to distinguish cured from uncured censored subjects. To take such feature into account, cure models were introduced by Boag (1949) and Berkson & Gage (1952).

Causal inference is a branch of statistics dedicated to the study of causal relationships between interventions and outcomes (Hernán & Robins 2020). It provides a theoretical and methodological framework for defining, identifying, and estimating causal effects. In clinical research, where establishing causal effects is of primary importance, this framework plays a central role. Over the past decades, this field has undergone substantial methodological advances to address the challenges involved in drawing causal conclusions from observational and experimental data.

Despite a well-developed literature on cure models (See Amico & Van Keilegom (2018) for a detailed literature review), little work has been done within the causal inference framework. Gao & Zheng (2017) were the first to propose two causal estimands for evaluating the effect of treatment on the cured proportion and on the survival of uncured patients in the presence of treatment non-compliance. Wang et al. (2023) developed an approach for estimating the causal effect of treatment on survival outcomes with a cure fraction by combining propensity-score stratification and covariate adjustment. More recently, Wang et al. (2024) used principal stratification (Frangakis & Rubin 2002) to define and estimate several causal estimands, including causal effects on the probability of being cured, survival at a fixed time point, and restricted mean survival time (Royston & Parmar 2011) for individuals who would be cured under either treatment condition. Despite recent advances in the definition of causal estimands for survival data with a cure fraction, the range of available methods for causal effect estimations remains scarce.

Given ongoing medical advances, cure is becoming an increasingly realistic outcome for many diseases, making cure models progressively more important. Therefore, there is a clear need to develop causal inference methods for survival data with a cure fraction, particularly in observational studies, whose use in clinical research continues to grow.

Mission

As postdoctoral researcher, you will develop causal inference methods, particularly for the estimation part, for survival data with a cure fraction, focusing on two causal estimands:

- The treatment effect on cure probability; and
- The treatment effect on survival.

The main objective will be to develop estimation methods based on propensity scores for observational data. The project will also involve the development of an R package and collaborations with the Montpellier University Hospital.

Profile

- PhD in Statistics in causal inference or survival analysis with a knowledge of the other field
- Fluent English – Level B2 or higher (Knowledge of French is a plus)
- Good R programming skills
- Writing skills
- Independent work
- Interpersonal skills

Supervision

Maïlis Amico, Junior Professor at the University of Montpellier
Website : maamico.github.io
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Location

Desbrest Institut of Epidemiology and Public Health (IDESP)
Campus Balard – Bâtiment 39
300 Avenue du Professeur Emile Jeanbrau
34090 Montpellier (France)

Duration

12 months, as soon as possible.

Salary

2 760 euros gross per month.

Application

Send a CV, a cover letter, a short description of research activities (1-2 pages) to Maïlis Amico (mailis.amico@umontpellier.fr)

References

Amico, M. & Van Keilegom, I. (2018), 'Cure models in survival analysis', *Annual Review of Statistics and Its Application* **5**, 311–342.

Berkson, J. & Gage, R. (1952), 'Survival curve for cancer patients following treatment', *Journal of the American Statistical Association* **47**, 501–515.

Boag, J. (1949), 'Maximum likelihood estimates of the proportion of patients cured by cancer therapy', *Journal of the Royal Statistical Society - Series B* **11**, 15–53.

Frangakis, C. & Rubin, D. (2002), 'Principal stratification in causal inference', *Biometrics* **58**, 21–29.

Gao, X. & Zheng, M. (2017), 'Estimating the causal effects in randomized trials for survival data with a cure fraction and non compliance', *Communication in Statistics - Theory and Methods* **46**, 4065–4087.

Hernán, M. & Robins, J. (2020), *Causal Inference: What If*, Chapman & Hall/CRC, Boca Raton.

Royston, P. & Parmar, M. (2011), 'The use of restricted mean survival time to estimate the treatment effect in randomized clinical trials when the proportional hazards assumption is in doubt', *Statistics in Medicine* **30**(19), 2409–2421.

Wang, Y., Deng, Y. & Zhou, X.-H. (2024), 'Causal inference for time-to-event data with a cured subpopulation', *Biometrics* **80**(2), ujae028.

Wang, Z., Wang, C. & Wang, X. (2023), 'Estimating causal effects in observational studies for survival data with a cure fraction using propensity score adjustment', *Biometrical journal* **65**, 2100357.